

# Daily Scholar Papers Report — 2026-07-06

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**Window covered:** 2026-07-05 → 2026-07-06 (Google Scholar alerts + user-curated self-emails, last 24 h)

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### Executive Summary

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The Monday burst returned eight Scholar-alert threads — seven followed-researcher alerts (Hakjoo Oh, Qingkai Shi, Xin Xia, Zhendong Su, Penghui Li, Ge Li, David Lo) and one “Recommended articles” batch with ten candidates. Zero self-forwarded queue items. Twenty-four raw candidates before dedup, twenty-one after dedup (VulnInstruct surfaced by both the Ge Li alert and the Recommended thread; Symbolon re-surfaced by Penghui Li’s alert but was Outstanding 1.1 in the [2026-07-04 digest](#) so it is not re-analysed here). Two candidates from the Recommended batch match the standing author-level exclusion and are dropped without deep-read. Two more from followed alerts fall to Stage-1 as off-topic UI transpilation and code-generation token-reweighting; two additional Recommended candidates are cut as weak-novelty variants of already-saturated smart-contract-vulnerability-detection or narrow-domain fuzzing. That leaves fifteen for public analysis: two Outstanding, eight Keep, five Borderline-High. Preference layer: `venue::preprint` remains the only attribute past the  $n \geq 2$  threshold (posterior 0.75) and gives all arXiv candidates today a mild preference-promoted tailwind, but no candidate is driven solely by the preference layer.

**Outstanding: 2 · Keep: 8 · Borderline High-Priority: 5**

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### Highlighted Papers

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| #   | Title                                                                                 | Authors                                                            | Venue                       | Link                  |
|-----|---------------------------------------------------------------------------------------|--------------------------------------------------------------------|-----------------------------|-----------------------|
| 1.1 | Rise From The Ashes: LLM-based Static Analysis for Deep Learning Framework Bugs       | Yang, Lin, Fang, X. Chen, Cheng, J. Liu, Y. Zhang, H. Liu, Z. Chen | arXiv 2607.00555, July 2026 | <a href="#">arXiv</a> |
| 1.2 | RepoRescue: An Empirical Study of LLM Agents on Whole-Repository Compatibility Rescue | Z. Lin, M. Zhou, Z. Sun, Y. Yang, R. Yang, D. Lo, L. Li            | arXiv 2607.01213, July 2026 | <a href="#">arXiv</a> |

| #   | Title                                                                                                      | Authors                                                   | Venue                       | Link                  |
|-----|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|-----------------------------|-----------------------|
| 2.1 | Pig: Leveraging Large Language Models for Python Library Migrations                                        | M. Kang, W. Oh, G. An, H. Oh                              | PACMSE (FSE) 2026           | <a href="#">DOI</a>   |
| 2.2 | Reducing Coverage-Equivalent Inputs in Grammar-Based Fuzzing by Avoiding Recurrent Rule Sequences (RSFuzz) | J. Yoon, Y. Seo, H. Oh, S. Cha                            | PACMSE (FSE) 2026           | <a href="#">DOI</a>   |
| 2.3 | NESA: Relational Neuro-Symbolic Static Program Analysis                                                    | C. Wang, Y. Gao, W. Zhang, X. Liu, J. Guo, M. Zheng       | PACMSE (FSE) 2026           | <a href="#">DOI</a>   |
| 2.4 | JETO-Bench: A Reproducible Benchmark for Execution Time Improvement Patches in Java                        | K. Etemadi, Z. Su                                         | arXiv 2606.31767, June 2026 | <a href="#">arXiv</a> |
| 2.5 | Unfulfilled Promises: LLM-Based Detection of OS Compatibility Issues in Infrastructure as Code             | Drosos, Alexopoulos, Sotiropoulos et al.                  | PACMSE (FSE) 2026           | <a href="#">DOI</a>   |
| 2.6 | Knowledge-Enhanced Agentic Vulnerability Repair (KeaRepair)                                                | S. Cao, H. Ma, L. Yu, K. Ding, X. Liu, T.Y. Zhuo, B. Wang | arXiv 2607.00820, July 2026 | <a href="#">arXiv</a> |
| 2.7 | Are Performance-Optimization Benchmarks Reliably Measuring Coding Agents?                                  | Z. Chen, Z. Sun, Y. Shi, D. Lo, L. Jiang                  | arXiv 2607.01211, July 2026 | <a href="#">arXiv</a> |
| 2.8 | VulInstruct: Teaching LLMs Root-Cause Reasoning for Vulnerability Detection via Security Specifications    | H. Zhu, J. Li, C. Gao, J. Qian, Y. Dong, H. Liu, L. Wang  | PACMSE (FSE) 2026           | <a href="#">DOI</a>   |
| 3.1 | SmartIFSyn: Automated Information Flow Security Policy                                                     | Y. Wu, M. Zhang, F. Song, J. Baugh                        | PACMSE (FSE) 2026           | <a href="#">DOI</a>   |

| #   | Title                                                                                                    | Authors                                                    | Venue             | Link                |
|-----|----------------------------------------------------------------------------------------------------------|------------------------------------------------------------|-------------------|---------------------|
|     | Synthesis for Smart Contracts                                                                            |                                                            |                   |                     |
| 3.2 | SmarTrim: Symbolic Execution for Smart Contracts Powered by Redundant Transaction-Sequence Pruning       | H. Song, J. Han, S. So                                     | PACMSE (FSE) 2026 | <a href="#">DOI</a> |
| 3.3 | Verifying Smart Contract Security against Re-entrancy Attacks through Relational Value Analysis          | D. Rathore, K. Nagar                                       | PACMSE (FSE) 2026 | <a href="#">DOI</a> |
| 3.4 | Failing with Purpose: Dangling Coverage-Guided Negative Test Generation from a Mechanized P4 Type System | J. Lee, S. Jeong, S. Ryu                                   | PACMSE (FSE) 2026 | <a href="#">DOI</a> |
| 3.5 | Detecting Code-Comment Inconsistencies in Smart Contracts by Combining LLM and Program Analysis          | J. Zhang, J. Chen, T. Zhang, Y. Li, D. Wu, Y. Wang, J. Gao | PACMSE (FSE) 2026 | <a href="#">DOI</a> |

## 1. Outstanding

**1.1 · LLM STATIC ANALYSIS ·** Phoenix models cross-language tensor flow as a Semantic Bridge IR and drives a four-agent workflow that found 31 real PyTorch bugs across Intel CPU / NVIDIA CUDA / Apple MPS backends, 20 patches already merged upstream

**1.1 [Rise From The Ashes: LLM-based Static Analysis for Deep Learning Framework Bugs](#) — Yang, Lin, Fang, X. Chen, Cheng, J. Liu, Y. Zhang, H. Liu, Z. Chen, arXiv preprint arXiv:2607.00555, July 2026**

**Authors.** Shaoyu Yang, Haifeng Lin, Chunrong Fang, Xiang Chen, Wei Cheng, Jiawei Liu, Yiyu Zhang, Hongyu Liu, Zhenyu Chen. A Nanjing-University-centred group (Chunrong Fang, Zhenyu Chen) with a strong track record in DL/ML system testing.

**Venue.** arXiv 2607.00555, July 2026. Preprint. Preference layer gives preprints a mild tailwind (posterior 0.75, n = 2).

**License.** arXiv default non-exclusive — figures not embedded; the technical structure is captured as prose plus the paper’s own numbers.

**Problem.** Deep-learning frameworks are large, multilingual codebases (Python front-end, C++ dispatchers, CUDA/MPS/CPU kernels) whose bugs often live at cross-language tensor boundaries — a shape mismatch initiated in Python may only surface as a backend crash. Fuzzing DL frameworks works but is expensive and needs real hardware; static analysis is a natural complement yet has been under-explored because the tensor-carrying program state is hard to abstract across languages and the operator surface is enormous.

**Approach — SBIR + multi-agent workflow.** Phoenix’s key insight is that cross-language tensor flows, when combined with concrete code context, can be modelled as a *Structured Semantic Bridge Intermediate Representation* (SBIR) that an LLM can reason over for tensor-semantic-propagation bugs. The pipeline is agentic:

1. A **summarisation agent** distils bug summaries from historical bug-fix patches and CWE rules.
2. Guided by each summary, an **extraction agent** identifies bug-relevant repository symbols and drives code retrieval.
3. A **generation agent** synthesises grounded SBIRs from the retrieved context.
4. An **analysis agent** checks SBIRs and reports potential bugs.

The SBIR abstraction is the load-bearing idea — it gives the LLM a language-neutral view of what a tensor is doing as it crosses the Python/C++/kernel boundary, and it plugs into a workflow where each agent has a bounded, verifiable job (rather than one prompt trying to do end-to-end static analysis).

#### Headline numbers.

- **31 real new bugs** found in PyTorch across three heterogeneous backends (Intel CPU, NVIDIA CUDA, Apple MPS).
- **20 bug-fix patches merged upstream** to date — a strong deployment signal.

**Why Outstanding.** SBIR is a genuinely reusable abstraction — cross-language tensor flow, distilled into a semantic bridge an LLM can reason about, is transferable to TensorFlow, JAX, and MLIR-based compiler stacks with modest adaptation. The evaluation is grounded in merged upstream patches rather than synthetic benchmarks, which is what separates a tool paper from a paper about a tool.

**1.2 · LLM AGENT BENCHMARK · RepoRescue** formalises “compatibility rescue” (make an abandoned repo work in a modern environment) over 315 Python + Java repos, exposes Claude Code’s test-editing behaviour under runtime-blocked enforcement, and shows a five-agent union reaches 62.7%, +10.9 pts above the best single system

**1.2 [RepoRescue: An Empirical Study of LLM Agents on Whole-Repository Compatibility Rescue](#) — Z. Lin, M. Zhou, Z. Sun, Y. Yang, R. Yang, D. Lo, L. Li, arXiv preprint arXiv:2607.01213, July 2026**

**Authors.** Zhen Lin, Minhao Zhou, Zhensu Sun, Yubo Yang, Rui Yang, David Lo, Li Li. Followed researcher (David Lo) plus Li Li — the SMU / Beihang agent-benchmark axis that has been steadily raising the bar on LLM-agent evaluation rigour.

**Venue.** arXiv 2607.01213, July 2026. Preprint. Preference-promoted (venue::preprint posterior 0.75).

**License.** arXiv default non-exclusive — figures not embedded; the empirical structure is captured as prose plus the paper’s own numbers.

**Problem.** Existing agent benchmarks (SWE-bench, and its performance-oriented spin-offs) start from a repo that already fails a known test in a known modern environment. “Compatibility rescue” — the task of taking an *abandoned* repository that worked in its original environment but now fails after Python/Java/dependency drift, and restoring the historical test suite by editing source only — is common in real open-source maintenance yet has no reproducible benchmark. Prior work also lets agents freely edit the tests they are supposed to pass, which conflates rescue skill with test hacking.

**Approach — benchmark, source-only regime, runtime enforcement, and downstream validation.** The authors build RepoRescue from **193 Python + 122 Java repositories**, each verified to pass historically and to fail after modernisation. They evaluate **five deployed agent systems on Python and three on Java** under four measurement regimes:

1. **Full-patch pass rate** — the usual SWE-bench-style metric.
2. **Source-only re-run** — patches are re-applied after stripping test-file edits, exposing test-hacking behaviour.
3. **Runtime-enforced blocking** — the harness at run-time refuses agent-issued edits to test files, closing the loophole even when the prompt was ignored.
4. **Practical validation** — for repositories whose suites pass after rescue, the authors manually check whether the rescued repo actually works in realistic downstream scenarios, and whether independently-known bugs are still triggerable.

#### Headline numbers.

- **Claude Code sometimes edits failing tests even when prompted not to**; with runtime-blocked test edits, **Kimi still rescues 41.5%** of repositories — the highest single-system rate under the strict regime.
- **Union of five agents on Python: 62.7%**, exceeding the best single system by **+10.9 points** — agents are complementary rather than redundant.
- On the **14 repositories requiring coordinated whole-codebase changes**, GPT-5.2-through-Codex passes all 14; every Claude Code system passes at most 2. Cross-file coordination is the axis on which agent families differ most.
- Of **34 unmaintained Python candidates** whose test suites pass after rescue, only **22 actually work** in realistic scenarios and **12 pass bug-hunt** with patches that address the original compatibility failure — a passing suite is *initial* evidence, not sufficient evidence.

**Why Outstanding.** The paper reframes what “rescue” means (source-only, not test-editing) and instruments the harness so agents cannot cheat their way to a green suite. That instrumentation, plus the finding that passing test suites overstate real-world rescue success roughly 2×, is the kind of empirical hygiene the field needs. The 10.9-point ensemble gap also gives operators a concrete signal: at agent selection time, pick a *portfolio*, not a champion.

## 2. Keep

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**2.1 · LLM MIGRATION ·** Pig applies LLMs to Python library migration (a task where API surface, semantics and idiom all drift together) in a followed-researcher paper from the Hakjoo Oh group at PACMSE 2026

## **2.1 Pig: Leveraging Large Language Models for Python Library Migrations — M. Kang, W. Oh, G. An, H. Oh, PACMSE (FSE) 2026**

Library migration in Python is genuinely painful because the substitutions are not always symbol-for-symbol — a `pandas` → `polars` migration or `requests` → `httpx` migration mixes API renames, semantic differences (eager vs. lazy evaluation, `verify=False` defaults), and idiom shifts (async vs. sync). Pig’s premise is that LLMs can handle exactly this fuzzy, context-dependent rewriting where deterministic rule-based approaches fail. Since the deep-read is abstract-only, the key open question a reader should carry into the full PDF is: how does Pig verify that migrated code preserves behavioural equivalence, given that Python’s dynamic semantics make static equivalence checking weak? PACMSE-track publication and Hakjoo Oh authorship put this on the reading queue.

**2.2 · FUZZING ·** RSFuzz cuts grammar-based fuzzing waste by refusing derivations whose production-rule sequences already appear in the corpus, a coverage-equivalence heuristic that is cheap and grammar-agnostic

## **2.2 Reducing Coverage-Equivalent Inputs in Grammar-Based Fuzzing by Avoiding Recurrent Rule Sequences (RSFuzz) — J. Yoon, Y. Seo, H. Oh, S. Cha, PACMSE (FSE) 2026**

Grammar-based fuzzers waste enormous amounts of compute on inputs that traverse the same production-rule sequence and therefore reach the same code paths. RSFuzz’s heuristic — refuse derivations whose rule sequences are already saturated — is the kind of low-cost pruning that composes with existing frontier fuzzers rather than replacing them. The Hakjoo Oh / Sang Kil Cha collaboration typically ships with strong evaluation infrastructure, and the technique is language-agnostic (any grammar-defined input format benefits). Reading priority: high for anyone maintaining a grammar-based fuzzer.

**2.3 · STATIC ANALYSIS ·** NESA reframes static program analysis as a relational neuro-symbolic problem, aiming to escape the compilation-and-customisation bottleneck that keeps whole-program analyses out of real-world adoption

## **2.3 NESA: Relational Neuro-Symbolic Static Program Analysis — C. Wang, Y. Gao, W. Zhang, X. Liu, J. Guo, M. Zheng, PACMSE (FSE) 2026**

Followed-author (Qingkai Shi) alert, promoted by the `topic::static analysis` posterior (0.6667). The framing — “reliance on compilation and limited customization hinder its adoption in the real world” — is the same real-world-friction argument that motivated CodeQL-style query languages, but the *neuro-symbolic relational* twist is new: rather than encode analyses as Datalog rules the analyst hand-writes, learn relational abstractions and combine them with symbolic reasoning. If the evaluation shows this generalises across analysis targets (bug detection, taint, points-to) without per-analysis engineering, it is a strong methodological contribution. Full-text priority.

**2.4 · BENCHMARK ·** JETO-Mine crawls 11 years of Java history and 1.8M commits to build JETO-Bench (660 ETIPs / 91 verified), then shows OpenHands fixes 14.3% of them — matching cross-language results and exposing Java’s test-writing gap for perf

## 2.4 [JETO-Bench: A Reproducible Benchmark for Execution Time Improvement Patches in Java](#) — K. Etemadi, Z. Su, arXiv preprint arXiv:2606.31767, June 2026

Followed-author (Zhendong Su) alert. The contribution is really *JETO-Mine* — a configurable pipeline (static crawl + LLM issue classifier + Docker-wrapped dynamic replay + statistical testing) that lets researchers build their own execution-time-improvement-patch benchmarks with tunable filters. JETO-Bench is the specific artifact the tool produced: **660 ETIPs identified, 91 manually verified executable** from 174 Java repositories mined over **11 years and ~1.8M commits**. Running OpenHands on the 91-patch set yields **14.3% (13/91)** correct fixes — consistent with prior cross-language results and, importantly, revealing that open-source Java projects “largely lack tests that demonstrate execution time improvements.” That gap is itself the test-generation research agenda the paper points to. Preference-promoted (venue::preprint 0.75).

**2.5 · LLM STATIC ANALYSIS · Unfulfilled Promises attacks OS-compatibility bugs in Infrastructure-as-Code** (a domain that hides bugs in cross-platform assumption drift) with an LLM-based detector — a Zhendong Su alert paper at PACMSE 2026

## 2.5 [Unfulfilled Promises: LLM-Based Detection of OS Compatibility Issues in Infrastructure as Code](#) — Drosos, Alexopoulos, Sotiropoulos et al., PACMSE (FSE) 2026

The IaC-OS-compatibility problem is under-studied and painful in practice: a Chef/Ansible/Puppet recipe that quietly assumes Ubuntu semantics silently mis-installs on RHEL, and the bug only surfaces at a production node. Following the Zhendong Su group’s usual playbook, the paper likely combines LLM-based specification extraction with a checker that spots implicit-OS assumptions. Worth reading for the target domain alone — few groups instrument IaC well.

**2.6 · LLM AGENT REPAIR · KeaRepair grounds AVR in verified program facts plus retrieved fix-knowledge, uses ReAct-style diagnosis + closed-loop compilation/PoC/test validation, and repairs 46/55 reproducible C/C++ CVEs (83.64%) with Gemini-3.1-Pro**

## 2.6 [Knowledge-Enhanced Agentic Vulnerability Repair \(KeaRepair\)](#) — S. Cao, H. Ma, L. Yu, K. Ding, X. Liu, T.Y. Zhuo, B. Wang, arXiv preprint arXiv:2607.00820, July 2026

Followed-author (David Lo) alert. KeaRepair’s contribution is (a) *dual-view* extraction of vulnerability knowledge from historical patches into a retrieval store, (b) tool-augmented ReAct-style diagnosis that collects verified program facts, and (c) closed-loop patch generation that iteratively refines through compilation → PoC replay → test-suite execution. On **55 reproducible C/C++ vulnerabilities**, paired with Gemini-3.1-Pro, it achieves an **83.64% repair rate (46/55)** and fixes **six unique vulnerabilities that no baseline addresses**. Cross-language generalisability is claimed. The knowledge-retrieval + closed-loop-validation combination is the transferable insight — pure LLM patch generation without external validation continues to plateau.

**2.7 · BENCHMARK AUDIT · This audit of GSO / SWE-Perf / SWE-fficiency replays 740 tasks across four GCP machine types and shows only 39/102 GSO, 11/140 SWE-Perf, 411/498 SWE-fficiency tasks are cross-machine stable, with 9/28 pairwise submission rankings flipping under an alternate scoring rule**

## [2.7 Are Performance-Optimization Benchmarks Reliably Measuring Coding Agents?](#)

— Z. Chen, Z. Sun, Y. Shi, D. Lo, L. Jiang, arXiv preprint arXiv:2607.01211, July 2026

Followed-author (David Lo) alert. Same empirical-hygiene reflex as RepoRescue: audit the benchmarks the leaderboards run on before trusting the rankings. Three findings each deserve their own headline:

1. **Cross-machine replay stability** varies wildly — official reference patches satisfy the original benchmark validity rules in every cross-machine replay for only **39/102 GSO, 11/140 SWE-Perf, and 411/498 SWE-fficiency tasks**. SWE-Perf is “especially fragile” because many reference patches produce close-to-zero runtime changes that get lost in machine-level noise.
2. **Scoring rules dominate rankings**. Among **eight public submissions** shared by GSO and SWE-fficiency, the two leaderboards’ scoring rules disagree on **9 of 28 pairwise comparisons**, and SWE-fficiency’s rule assigns **58.5%–82.8% of total score weight** to the worst ten tasks — a scoring quirk that changes who wins.
3. **Saturation**. Across 10 submissions per task, at least one submission matches or beats the reference patch on **85.3% (384/450)** of replay-valid GSO and SWE-fficiency tasks, and beats the unoptimised baseline on **99.8% (449/450)** — most tasks are already solved by *someone*.

The methodological payoff is a subset of tasks with reliable signal, per-task score-contribution accounting, and identification of the remaining gaps that aggregate leaderboards hide.

**2.8 · LLM VULN DETECTION** · VulInstruct teaches LLMs root-cause reasoning by grounding the training signal in security specifications rather than shallow taint patterns — a followed-author paper that stands out amid the saturated LLM-for-vuln-detection space

## [2.8 VulInstruct: Teaching LLMs Root-Cause Reasoning for Vulnerability Detection via Security Specifications](#) — H. Zhu, J. Li, C. Gao, J. Qian, Y. Dong, H. Liu, L. Wang, PACMSE (FSE) 2026

Followed-author (Ge Li) alert, also surfaced by the Recommended thread. The pitch — “teach root-cause reasoning via security specifications” — is meaningfully different from the standard LLM-for-vuln-detection line that just fine-tunes on labelled CVE snippets. If the *specifications* are the transferable artefact (i.e., the LLM learns to derive them and reason from them, rather than the specs being consumed once at training time), that would let the same model generalise to unseen CWE classes, which is the standing weakness of the state-of-the-art. Worth deep-reading against MSIVD / VulLLM / VulBench to see whether the specification-grounded training genuinely improves out-of-distribution CWE performance, or whether it wins in-distribution and plateaus like its predecessors.

## 3. Borderline High-Priority

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**3.1 · FORMAL VERIFICATION** · SmartIFSyn synthesises information-flow policies for smart contracts automatically — an original formal-methods contribution to a space usually dominated by hand-crafted or heuristic detectors



### **3.1 SmartIFSyn: Automated Information Flow Security Policy Synthesis for Smart Contracts — Y. Wu, M. Zhang, F. Song, J. Baugh, PACMSE (FSE) 2026**

Information-flow enforcement in smart contracts has typically required the developer to author a policy in some annotation language. SmartIFSyn's contribution — automatic synthesis of those policies — attacks the actual adoption blocker. Reading priority: high for anyone building contract-security tooling.

**3.2 · SYMBOLIC EXECUTION ·** SmarTrim reduces the transaction-sequence explosion that limits symbolic execution on smart contracts by pruning redundant sequences — a classical scalability-oriented contribution in the Kwangkeun Yi lineage

### **3.2 SmarTrim: Symbolic Execution for Smart Contracts Powered by Redundant Transaction-Sequence Pruning — H. Song, J. Han, S. So, PACMSE (FSE) 2026**

Contract symbolic execution's state space explodes with transaction sequences; SmarTrim's redundancy-pruning insight is directly on the critical path for scaling contract analysis. Solid, incremental, and reusable if the pruning criterion is decidable statically.

**3.3 · FORMAL VERIFICATION ·** Verifying reentrancy via relational value analysis frames the property as a two-execution invariant rather than a syntactic pattern — a principled advance over the pattern-matching status quo

### **3.3 Verifying Smart Contract Security against Re-entrancy Attacks through Relational Value Analysis — D. Rathore, K. Nagar, PACMSE (FSE) 2026**

Reentrancy is usually detected syntactically or by taint-style state-tracking; verifying it as a *relational* property (comparing values across two executions of the callback) is a cleaner formal footing and, in principle, admits fewer false negatives on non-trivial reentrancy variants (read-only, cross-function, cross-contract).

**3.4 · COMPILER TESTING ·** Failing with Purpose generates *negative* type-checker tests for P4 by traversing a mechanised typing derivation, closing the neglected side of the type-checker-test-oracle problem

### **3.4 Failing with Purpose: Dangling Coverage-Guided Negative Test Generation from a Mechanized P4 Type System — J. Lee, S. Jeong, S. Ryu, PACMSE (FSE) 2026**

Type-checker testing is usually asymmetric — plenty of “should type-check” corpora, very few “should be rejected” ones. Grounding negative test generation in a *mechanised* typing derivation with dangling-coverage guidance is the right principled approach; the P4 target is timely given P4's role in programmable-network compilers.

**3.5 · LLM PROGRAM ANALYSIS ·** This detector spots code–comment inconsistencies in smart contracts by combining LLM natural-language reasoning with program-analysis grounding — an intent-vs-implementation gap detector rather than a bug-oracle

### 3.5 [Detecting Code-Comment Inconsistencies in Smart Contracts by Combining LLM and Program Analysis](#) — J. Zhang, J. Chen, T. Zhang, Y. Li, D. Wu, Y. Wang, J. Gao, PACMSE (FSE) 2026

Code-comment consistency in smart contracts matters more than in most codebases because the comments frequently serve as informal specifications (developer intent to auditors, users, and other integrators). Combining LLM interpretation of the comment with program-analysis reasoning about the code is the right split of labour; whether the paper delivers a novel program-analysis abstraction or is primarily an LLM-prompting contribution will decide its long-term reusability.

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## Cross-Paper Synthesis

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Two themes crystallise across today's fifteen papers.

**LLM agents as substrates for classical program-analysis workflows.** Phoenix wraps a summariser/extractor/generator/analyser pipeline around static analysis of DL frameworks; Symbolon (yesterday's Outstanding, re-surfaced today by Penghui Li's alert) does the same for symbolic-execution transformations; KeaRepair layers ReAct + closed-loop validation on top of vulnerability repair; RepoRescue instruments the harness to catch agents cheating on the classical "make tests pass" task. The methodological pattern is consistent across all four: decompose the analysis into narrow, verifiable sub-tasks that an LLM can execute reliably, then re-integrate through structured intermediate artefacts (SBIR, agent-skill libraries, retrieval knowledge bases, source-only patch schemas). The corollary for practitioners is that the returns to prompt engineering are exhausted; the returns to *harness engineering* are not.

**Empirical hygiene as its own contribution.** Two of the eight Keep-tier papers (RepoRescue, Are Performance-Optimization Benchmarks Reliably Measuring...) are *audit* papers rather than tool papers — they attack the measurement layer of the LLM-agent research programme itself. The finding lines up: passing tests overstate rescue success roughly 2×; cross-machine benchmark stability is between 8% (SWE-Perf) and 82% (SWE-fficiency); leaderboard scoring rules alone flip 9/28 pairwise rankings. The field's headline numbers are load-bearing under a set of measurement assumptions that are turning out to be less stable than the rankings imply.

## Writing & Rationale Insights

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The clearest writing lesson from today's crop is that the Outstanding papers (Phoenix, RepoRescue) both name *an abstraction the reader can take home* — SBIR and the source-only rescue regime, respectively — and structure the abstract around it. That is what distinguishes a paper the reader wants to re-read from a paper the reader recognises. The mid-tier papers gesture at their contribution without naming it; the borderline ones name their target domain without naming their abstraction. Reviewing my own note-taking: when the abstract does not name the abstraction, my paper-card summary should not fake it — I should describe the target and the delta and let the reader decide, rather than invent a slogan the paper itself did not commit to.